

YUSEF SYED

Toronto, ON | U.S.-Canadian dual citizen

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Education

University of Toronto

Expected May 2030

Mathematics and Computer Science (Intended Double Major)

Faculty of Arts & Science, first-year Mathematical & Physical Sciences

Current coursework: programming, calculus with proofs, linear algebra, statistical reasoning. Prior coursework: Data Structures, Lehigh Carbon Community College (dual enrollment).

Research Projects

Agent Eval Mutation Lab | [Code and evidence](#)

Aug 2026 - Present

Python, Inspect, Docker, SQLite

- Built a resumable evaluation engine separating proposed actions, executed tool effects and unknown outcomes; **104 scorer configurations across 13 synthetic scenarios**.
- Integrated Docker-isolated Inspect tools with independent database-effect checks; tested commit-before-error and later-revocation faults, plus **24 local-model tool-use samples**.
- Preregistered a 624-trial two-model study; reported **5 of 6 failed validity gates**, withheld comparative model claims and documented missing-output sensitivity.

Tiraz Garment Completion Study | [Code and results](#)

Aug 2026

Python, PyTorch, NumPy

- Trained three seeded set encoders on **27,683 Fashionpedia annotation groups**; achieved **58.5-59.0%** top-1 versus a **52.7%** co-occurrence baseline on 962 held-out source groups.
- Separated training, tuning and calibration groups; context removal reduced nominal 90% prediction-set coverage to **55.6-57.9%**. Annotation-only study, not image recognition.

ShiftProof | [Code and evaluation](#)

Sep 2026

Python, local LLM, constraint verification

- Built a local scheduling prototype with independent constraint verification and coordinator approval; validated a synthetic workflow and audited four model requests for raw-name and note exclusion.

Open-Source Contributions

Contributor | [Merged contributions](#)

Aug 2026 - Present

- Contributed **9 merged upstream PRs** across Microsoft Agent Lightning, Microsoft PyRIT, W&B RAI Toolkit, Meridian Labs Inspect Scout and Kornia.
- Fixed asynchronous child scoring and unassessed-result handling in W&B's [CompositeScorer](#); added focused regressions for aggregation and async behavior.
- **Open PyTorch PR:** implemented CPU/CUDA shape gradients for regularized incomplete gamma. Local source-build checks compared **3,486 derivative outputs** across two CPU builds and RTX A4000 CUDA, with no corpus failures. [PR](#) | [Validation](#)

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL

Research and testing: PyTorch, NumPy, Inspect, pytest, mypy, Git, GitHub Actions

Systems: Docker, SQLite, PostgreSQL; C++/CUDA contribution experience

Leadership

Feeding the Valley: Vice President (2025-2026) | **FIRST Robotics Team 834:** Fundraising Lead (2022-2026)

AI-assisted development; contribution details, methods and validation are linked.